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Крайне негативные эмоции мешают когнитивной деятельности: данные о вызванных потенциалах и динамических нейронных сетях мозга

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Основные моменты

- В последние годы взаимосвязь между эмоциями и когнитивными процессами была одной из самых обсуждаемых тем. Однако до сих пор неясно, какие именно эмоции могут существенно влиять на когнитивные процессы и каким образом. В этом исследовании мы разработали новую парадигму эксперимента «Аффективный Струоп» для изучения этих вопросов.
- Крайне негативные (EN), умеренно негативные (MN), умеренно позитивные (MP), чрезвычайно позитивные (EP) и нейтральные изображения были показаны перед заданиями Stroop. Поведенческие результаты показали, что EN-эмоция значительно влияла на когнитивные способности по сравнению с другими типами эмоций, со значительным

увеличением времени реакции в состоянии EN-эмоции ($P<0{,}05$).

- Кроме того, динамические механизмы работы мозга были проанализированы с точки зрения как вызванных потенциалов (ВП), так и изменяющихся во времени мозговых сетей. Результаты показали, что эмоция «удивление» вызывала более выраженные амплитуды N2, P3 и LPP в лобных, теменных и затылочных долях мозга. В то же время при выполнении теста Струпа в состоянии «удивления» амплитуды N2, P3 и LPP были меньше, чем в нейтральном состоянии. Это указывает на то, что эмоция «удивление» была приоритетной и требовала больше когнитивных ресурсов по сравнению с нейтральной эмоцией.

- На стадиях P3 и LPP мы наблюдали усиление восходящих связей между теменными и лобными долями при обработке эмоций, связанных с негативными событиями. Кроме того, при выполнении задачи Струпа в условиях негативных событий наблюдалась более сильная нисходящая связь между лобными и затылочными долями, обеспечивающая когнитивный контроль. Эти результаты свидетельствуют о том, что эмоции, связанные с негативными событиями, мешают когнитивной деятельности, потребляя больше когнитивных ресурсов, и мозгу приходится усиливать когнитивный контроль, чтобы справиться с задачей Струпа.

- Эти результаты могут быть полезны для распознавания и регуляции эмоций.

Абстрактный

In recent years, the relationship between emotion and cognition was a hot topic. However, it remains unclear which specific emotions can significantly interfere with cognition and how they do so. In this study, we designed a novel Affective Stroop experiment paradigm to investigate these issues. The extremely negative (EN), moderately negative (MN), moderately positive (MP), extremely positive (EP) and neutral pictures were displayed before Stroop tasks. The behavioral results revealed that EN emotion significantly interfered with cognitive performance compared to other types of emotions, with a significant increase in reaction time under the EN emotion condition ($P<0.05$). Furthermore, the dynamic brain mechanisms were analyzed from both Event-Related Potential (ERP) and time-varying brain network perspectives. Results showed that EN emotion evoked larger N2, P3, and LPP amplitudes in the frontal, parietal, and occipital brain regions. In contrast, the Stroop task under EN condition led to smaller N2, P3, and LPP amplitudes compared to neutral condition. This indicates that EN emotion was prioritized and consumed more cognitive resources relative to neutral emotion. During the P3 and LPP stages, we observed enhanced bottom-up connections between the parietal and frontal regions while the processing of EN emotion. Additionally, there were stronger top-down cognitive control connections from the frontal to the occipital regions while processing the Stroop task under EN condition. These findings consistently suggest that EN emotion interferes with cognition by consuming more cognitive resources, and the brain needs to enhance cognitive control to support Stroop task execution.

Introduction

Studies have shown that human emotional states influence the performance of cognitive tasks (Dolcos, 2006, Vuilleumier et al., 2001). The spatial cognition and decision-making are important cognitive functions for pilots and bus drivers, if their cognitive performance is impaired by emotion, it will cause serious consequences (Eom and Han, 2012, Kaur, 2019). However, it is unclear what types of emotions can interfere with these cognitive functions and how specific emotions interfere with cognitive task processing. Investigating these issues is important to recognize and warn about certain emotions that interfere with cognition.

Behavioral results of many classical cognitive tasks have shown that negative emotions can impair cognitive performance. For instance, in a working memory task, negative emotional stimuli reduced the task performance compared to neutral stimuli(Dolcos, 2006; Uher et al., 2014). In the Affective Stroop task, aversion information could lead to slower responses (Monika et al., 2008), and negative emotional stimuli may also decrease the accuracy (Long et al., 2013). The similar impact of emotional stimuli was observed in N-back cognitive tasks, and the results showed that the presence of negative emotional stimuli can reduce accuracy during execution and increase reaction time (Uher et al., 2014). Positive, neutral, and negative emotional stimuli were shown to subjects during Simon task, and subjects performed worse in trials under negative emotion condition relative to other conditions (Monika et al., 2008). Although previous studies indicated that negative emotions may increase reaction time or reduce accuracy in most cognitive tasks. According to Lange's valence-arousal emotion model(Bradley and Lang, 2007), the emotion valence varying from low to high implies the emotion changing from negative to positive, i.e. EN emotion with most lower valence, MP emotion with lower valence, neutral emotion with middle valence, MP emotion with higher valence and EP emotion with highest valence. Positive and neutral emotions mostly bring no harm to our daily life. However, the valence of negative emotions is important, as negative emotions with different valence can bring different feelings and impacts on cognition. Researches have pointed out that EN emotion not only threaten the physical and mental health of human, but also affect their cognitive processing, such as can not make any response or led to unwise decision-making under extreme fear(Long et al., 2013; Huang and Luo, 2006). Emotional pictures with different valences were utilized as stimuli to investigate whether humans are sensitive to valence differences. Results showed that humans are sensitive to EN emotional stimuli, and the brain preferentially processes EN stimuli (Yuan et al., 2007). The influence of different levels of sadness on attention is also different, low-level sadness expands the scope of attention, while high-level sadness narrows the scope of attention(Hailu et al., 2018).

Event-related potential (ERP) can reflect brain activity at the millisecond level and has been widely used in research on emotion and cognitive processing (Li et al., 2016; Artyom et al., 2015). During a cognitive conflict experiment, the researchers detected distinct variances in the ERP components around 200ms after the stimuli were presented, with larger N200 amplitudes observed when dealing with negative emotional conflicts (Kanske and Kotz, 2010). In addition, many studies have pointed out that the N2 component of the anterior central brain region reflects the conflict processing process (Heidlmayr et al., 2020), and the amplitude of the N2 component represents cognitive control. A larger N2 amplitude indicates stronger cognitive control (Kanske and Kotz, 2010, Ladouceur et al., 2010). Some late ERP components such as P3 and late positive potential (LPP) have also been reported as important to emotional and cognitive processing, especially the LPP component, which is considered to represent emotion intensity and cognitive resource consumption (He et al., 2020, Brouwer et al., 2017, Enriquez-Geppert et al., 2010). The LPP component of the parietal region has been demonstrated that reflect emotion intensity, and the decreased LPP amplitude reflected a decrease in negative emotion intensity (Yueyao et al., 2023). Moreover, the amplitudes of LPP in the parietal and frontal regions were used as an index to evaluate the effect of negative emotion regulation (Zhao et al., 2021, Xie et al., 2022).

The important brain regions related to emotion and cognitive interactions have also been investigated based on EEG and fMRI. The parietal and prefrontal lobes are the most reported brain regions related to emotion and cognition interactions, and have been identified as sharing brain regions of emotion and cognition (Pessoa, 2008). For instance, the prefrontal cortex played an important role in emotion affect cognitive control experiment, greater activation of the prefrontal cortex was observed under negative emotions, while the activation of the prefrontal cortex was reduced in cognitive control tasks (Dichter et al., 2009). Besides the prefrontal region, multi-brain regions participated in emotion and cognition interactions. Specifically, in the Affective Stroop task based on fMRI, the amygdala, insula, and prefrontal cortex were activated, and the cognitive networks including the prefrontal cortex, superior temporal lobe, and insula were also activated (Raschle et al., 2017). The cognitive function of route planning during navigation needed the joint coding of multiple brain regions such as the hippocampus, the entorhinal cortex, and the forehead (Epstein et al., 2017). The prefrontal cortex and the left amygdala were involved in cognition and emotion processing, especially cognitive and emotion reappraisal (Scheffel et al., 2019). From the above, we can find that the processing of emotion and cognition, as well as the modulation of emotion on cognition, involves the activation of multiple brain regions and the cooperation of multiple brain regions. Brain network can represent the interaction and information exchange between multiple brain regions, and has been widely used in various fields in recent years. The long-range connections between the frontal and occipital lobes were demonstrated that can better represent the differences among emotions (Li et al., 2019). Our previous research also found that the long-range connections among the frontal, occipital lobes, and temporal lobes, are closely related to the differences in emotional processing (Yang et al., 2020). In Chinese language audio-visual conflict cognition tasks, the dense connections between the posterior occipital and parietal lobes were demonstrated helpful for prior processing of auditory conflict processing (Pei et al., 2022b, Pei et al., 2022a). EEG is characterized by its high temporal resolution and has thus been widely used to reveal the dynamic network topology during various cognitive processes, including language, attention, and motor imagery (Li et al., 2016). Therefore, utilizing time-varying brain networks may depict the dynamic network patterns of emotion affect cognition.

In recent years, the Affective Stroop task is a classical paradigm used in many emotion-cognition interaction studies(Flaisch et al., 2019; Dragan et al., 2022). The Number Stroop experiment is the adaptation of Affective Stroop, and it contains the presentation of emotional pictures and numerical discrimination task. Previous studies have used the Number Stroop experiment paradigm to explore the activation characteristics of brain regions in emotion and cognition processing, as well as, the two-way modulation between emotion and cognition (Raschle et al., 2017; Blair et al., 2007). The Affective Number Stroop task involves many cognition function such as, emotion processing, spatial cognition, decision-making, and these cognition functions are important for pilots and bus drivers(Dragan et al., 2022). However, the previous Affective Stroop task did not consider the impact of different valence emotions on Stroop tasks. This study designed a new Affective Stroop experiment paradigm utilizing five different valence emotional pictures as stimuli to investigate which valence emotion has significant influence on cognition.

The purpose of this study is to explore what types of emotions can significantly interfere with spatial cognition, decision-making and the brain activation of specific emotions interfere with cognitive task processing. For this purpose, a new Affective Stroop experimental paradigm was designed to investigate the effects of different valence emotions on cognitive tasks. Then the ERPs and time-varying brain network were analyzed to uncover the dynamic brain mechanism of EN emotion impairing cognition.

Section snippets

Subjects

Forty healthy right-handed local college students (20 female, aged 18–28, mean age: 22 years) participated in the experiment. To ensure that the mental state of all subjects is normal, each subject has passed the Beck Depression Scale(Jackson-Koku, 2016) and the Beck Anxiety Scale (Fydrich et al., 1992), before the experiment no subjects had not taken medication that affected their mental status. Subjects had a normal or corrected normal vision. Before the experiment, each subject was informed …

Behavioral performance

The mean RTs of the Stroop congruent and incongruent under five valence emotional conditions were shown in Table 1. We first examined the effect of task and different valence emotions on RTs. There was a significant main effect of the task. Under each valence emotion condition, subjects were significantly ($F(1, 78) = 6.54$, $P < 0.05$) slower to respond to incongruent, relative to congruent trials. There was also a significant main effect of emotions. Subjects were significantly (F (1, 78) …

Discussion

This study explored the impact of different valence emotions on cognitive processing based on a new Affective Stroop Number experiment paradigm. The behavioral results indicated that EN emotion impaired cognitive performance, and subjects had significantly slower RTs under EN emotion condition relative to other emotion conditions. The ERP analysis and time-varying brain network analysis depicted the brain mechanism of EN emotion interfered with cognition. …

Conclusion

In this paper, we designed a new Affective Stroop experiment to examine the impact of different valence emotions on cognitive performance and the dynamic mechanism of the interference of emotion on cognition. The behavioral results confirm that EN emotions interfere with cognitive performance, and subjects respond significantly slower under EN emotion condition ($P < 0.05$). This means that EN emotion should be paid more attention in our daily life. The analysis results of ERP and brain network …

Author contribution statement

Kai Yang: Research design, Data collection, Data analysis, Manuscript writing. **Ying Zeng:** Research design, Data analysis, Manuscript writing. **Li Tong:** Research design, Data analysis, Manuscript writing. **Yidong Hu:** Data collection, Document retrieval. **Rongkai Zhang:** Research design, Data analysis. **Zhongrui Li:** research design. **Bin Yan:** Research design, Manuscript writing. …

CRediT authorship contribution statement

Zhang Rongkai: Methodology, Formal analysis. **Li Zhongrui:** Investigation, Data curation. **Yan Bin:** Writing – review & editing, Resources, Conceptualization. **Yang Kai:** Writing – original draft, Investigation, Formal analysis, Conceptualization. **Zeng Ying:** Writing – review & editing, Funding acquisition, Formal analysis, Data curation. **Tong Li:** Writing – review & editing, Methodology, Conceptualization. **Hu Yidong:** Writing – review & editing, Data curation. …

Declaration of Competing Interest

The authors report no declarations of interest. …

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> *Citation Excerpt :*
> …N2 and P3 are neural markers of attention allocation and cognitive control (Ahumada-Méndez et al., 2022; Gupta et al., 2019; Schupp et al., 2007). The N2 peaks over the frontal sites approximately 200–400 ms following stimulus onset and has been demonstrated to be sensitive to emotional valence (Yuan et al., 2007) and arousal (Lusk et al., 2015; H. Wu et al., 2014; Yang et al., 2023). The P3 peaks at frontal and parietal midline electrodes between 300 and 500 ms after stimulus onset.…

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